Project Title: Custom Surgical Grasper with Integrated Sensing for da Vinci Surgical Robot

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Problem Statement

Surgeons using the da Vinci surgical system for robotic minimally invasive surgery (RMIS) rely exclusively on stereoscopic visual feedback, lacking one of the most critical senses available to humans: touch (Othman et al., 2022). Without sensory feedback, operators risk applying extraneous force, causing tissue trauma and ischemic crushing (Burkhard et al., 2018; Hosseinabadi et al., 2021). Moreover, the inability to palpate makes identifying vital sub-structures, like blood vessels, highly dependent on experience (Chaturvedi et al., 2018). Current instruments lack onboard sensory capabilities, prolonging operations, and dramatically steepening the learning curve.

Previous Work

A. Background

Conventionally, during open surgery, surgeons had direct contact with the tissue of patients (Othman et al., 2022). However, the introduction of RMIS has physically separated surgeons from patients, unfortunately eliminating the sense of touch that was previously available in the process. Since then, surgeons have relied solely on visual cues from a screen, which may lead them to apply excessive force in the hope of preventing tissue slip; conversely, surgeons may attempt to hold tissue loosely to prevent ischemia, but that may result in tissue slip. Nevertheless, RMIS has introduced numerous benefits, including shorter hospital stays (Othman et al., 2022). That is why it is important to improve the experience of RMIS by adding sensor integration: more specifically, sensor integration maintains the numerous benefits of RMIS while reducing the annoyances associated with sensory deprivation.

Previous research has demonstrated the viability of developing modular, open-platform adaptors for da Vinci Si instruments to integrate with collaborative robotic systems. Researchers have successfully utilised 3D-printed PLA components to create custom mounts and couplings, proving that cost-effective prototyping can reliably withstand the applied forces required for tool manipulation and actuation (Habeeb et al., 2025).

B. Sensors

Other researchers have made attempts to integrate miniature sensors. Firstly, electrical sensors translate mechanical signals into electrical signals and include piezoresistive, capacitive, and magnetic (Hall-effect) technologies, which can be integrated directly into the jaws of surgical graspers (Qiu et al., 2024). These sensors can measure contact pressure, shear forces, and tissue stiffness. In fact, DeLorey et al. (2026) used Hall-effect sensors embedded in an elastomer to detect incipient slip. Next, Optical approaches include Bragg Grating and light intensity modulation to calculate force based on mechanical changes to light transmitted through optical fibers (Guo et al., 2025). These sensors are highly flexible, biocompatible, and immune to electromagnetic interference. Interestingly, sensorless approaches have also been studied to some degree. For example, instead of adding physical hardware, sensorless methods use existing robot data to estimate interaction forces. This may include model-based approaches such as motor currents, joint encoders, and vision-based methods that process the surgical stereoscopic video feed using artificial neural networks to estimate tissue deformation and interaction forces. Lastly, newer research directions point to applications using soft microfluidic and image-based sensors. Microfluidic sensors embed conductive liquid metals, such as Galinstan, in highly flexible elastomers, while image-based sensors use miniature cameras to track the physical deformation of a soft translucent membrane upon contact with tissue.

C. Limitations

While some of the aforementioned approaches have been promising, with the authors discussing their strengths and weaknesses, the solutions frequently suffer from complex electrical routing, bulkiness, or prohibitive micromachining costs that hinder practical deployment (Guo et al., 2025; Othman et al., 2022). Additionally, DeLorey et al. (2026) noted that when they used an elastomer with a smooth surface, the gripping ability of the grasper was significantly reduced. Also, it was noted that sensorless approaches are laggy for real-time experiences. This is because they use traditional deep learning models, which, while great for estimating force or detecting tissue slip, require computationally expensive processing. This introduces delay in processing that makes real-time feedback impractical for deployment in the operating theatre (Hosseinabadi et al., 2021). Considering all this evidence, the literature analysis shows a lack of a scalable, low-cost proof-of-concept grasper that embeds sensors without compromising the compact footprint. A compelling approach is to study one involves measurement of different sensory features to improve the surgical experience when performing RMIS. However, achieving multimodality remains a significant challenge because robotic end-effectors have severely limited space to accommodate multiplexing devices and complex electrical interfacing circuitry (Burkhard et al., 2018; Qiu et al., 2024).

Proposed Solution

The objective is to engineer a proof-of-concept multimodal sensor assembly for the da Vinci surgical grasper that integrates (1) near-infrared transillumination for vascular perfusion detection and (2) electrical bioimpedance for tissue classification.

References

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